1.49 Representing Data

Worked Solutions

1)

$\bar{x} = \frac{75}{14} = 5.36$ (3 SF)

2 a)

Unimodal, discrete, symmetric

2 b)

Unimodal, continuous, symmetric

2 c)

Unimodal, discrete, negatively skewed

2 d)

Bimodal, continuous, symmetric

2 e)

Unimodal, continuous, positively skewed

2 f)

Bimodal, discrete, negatively skewed

2 g)

Uniform, unimodal, discrete, symmetric

3 a)

Positive skew

3 b)

$Q_2 = 40$

3 c)

$Q_1 = 393.5$, $Q_3 = 412.5$ IQR= 412.5 −393.5 = 19

4 a)

$Q_2 = 86$ cm $Q_1 = 77$ cm, $Q_3 = 95$ cm IQR= 95 −77 = 18 cm

4 b)

Mean is $\bar{x} = 88$ cm (3 SF) Sample standard deviation is s= 15.1 cm (3 SF) (Standard deviation is 14.5 cm (3 SF))

4 c)

The median would be a better measure of central tendency as it is not affected by the tallest kangaroo at 129 cm, which is considerably taller than the next largest at 100 cm.

5 a)

Negative skew

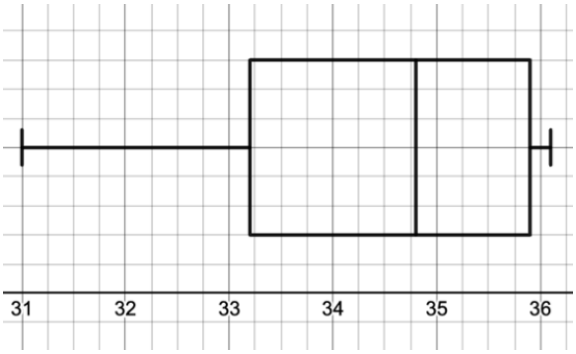
5 b)

36.0 is the mode

5 c)

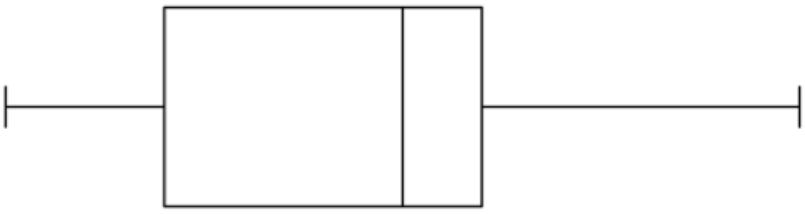
34.8 is the median

5 d)



6)

Any example where $Q_3 - Q_2 < Q_2 - Q_1$ e.g.



7 a)

Positive skew

7 b)

Negative skew

7 c)

Roughly symmetric

7 d)

Positive skew

8 a)

The Median = 15

8 b)

Positive skew

9 a)

$Q_2 = 14.5$

9 b)

Range = $30 - 7 = 23$

9 c)

$Q_2 = 12.5$, $Q_4 = 18.5$ $IQR = 18.5 - 12.5 = 6$

10 a)

Median of Class A = 66 Median of Class B = 62

10 b)

Class A: $Q_1 = 37.5$, $Q_3 = 92$ $IQR = 92 - 37.5 = 54.5$ Class B: $Q_1 = 34$, $Q_3 = 78$ $IQR = 78 - 34 = 44$

10 c)

Class A has a higher median at 66 than Class B at 62, so on average Class A performed better than Class B. Class A has a larger interquartile range at 54.5 than Class B at 44, so the scores of Class A are more spread out.

11)

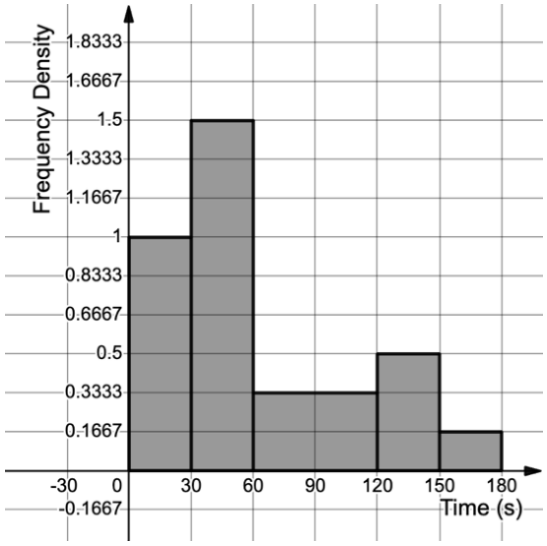
Area for $9 \leq x \leq 25$ is $3 \times 1.8 + 4 \times 1.4 + 9 \times 0.6 = 16.4$

Total area is $4 \times 0.8 + 8 \times 1.8 + 4 \times 1.4 + 12 \times 0.6 = 30.4$

The proportion is $\frac{16.4}{30.4} = \frac{41}{76} = 53.9\%$ (3 SF)

12 a)

Meeting Length (t)	Frequency
$0 \leq t < 30$	30
$30 \leq t < 60$	45
$60 \leq t < 120$	20
$120 \leq t < 150$	15
$150 \leq t < 180$	5



12 b)

Number of meetings that last between 40 and 99 minutes $= 20 \times 1.5 + 39 \times \frac{1}{3} = 43$

13 a)

Total Frequency $= 20 \times 3 + 10 \times 4 + 10 \times 1.5 + 10 \times 1 = 125$

The median is in the $\frac{125}{2} = 62.5$th position.

So the median is in the 30 to 40 class interval.

13 b)

Time (s) Midpoints	Frequency
20	$3 \times 10 = 30$
35	$4 \times 10 = 40$
45	$1.5 \times 10 = 15$
55	$1 \times 10 = 10$

Mean $\bar{x} = 33.9$ (3 SF)

13 c)

Sample standard deviation s= 11.4 (3 SF) (Standard deviation σ= 11.3 (3 SF))

13 d)

$\text{P}(28 < \text{Time} < 32) = \frac{2\times3+2\times4}{10\times3+10\times4+10\times1.5+10\times1} = \frac{14}{95}$

14)

$2 \times 6 = 12$ cm^2 represents 24 apples, so every 1 cm^2 represents 2 apples.

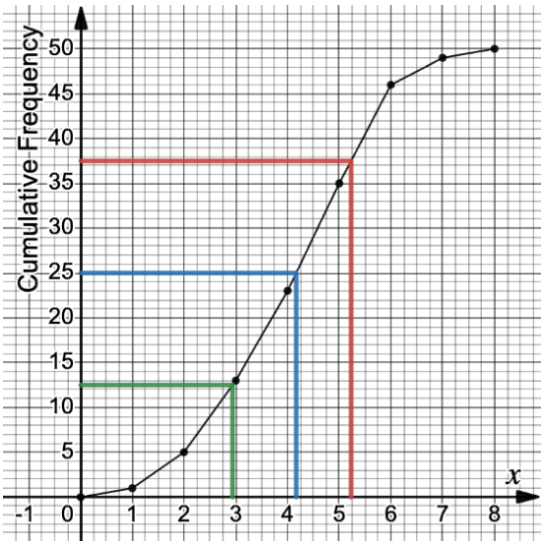
$\therefore$ 16 apples must be an area of 8 cm^2

160 to 180, a gap of 20 has a width of 2 cm

So 180 to 210, a gap of 30 must have a width of 3 cm

$\therefore 8 = h \times 3 \Rightarrow$ the bar must have a height of $\frac{8}{3}$ cm

15)



$$Q_1 = \frac{47}{16} = 2.94 \text{ (3 SF)}$$

$$Q_2 = \frac{25}{6} = 4.17 \text{ (3 SF)}$$

$$Q_3 = \frac{115}{22} = 5.23 \text{ (3 SF)}$$

16 a)

Graph B should be used as the data should be plotted at the top end of each class interval.

16 b)

The data is cumulative, so an increasing gradient would show that sales week on week are increasing. However, at the end of the 5th week, the gradient starts to decrease and so the number of sales here are decreasing.

16 c)

Amber only sold ice creams on a Sunday. Estimating from the graph would assume that the sales of ice creams were evenly spread across the week, which is not the case.